RUNE: Modular Training for Machine-Learned Interatomic Potentials

Technical report · liujiahui@zgci.ac.cn

Summary

RUNE is a modular package for training and deploying machine-learned interatomic potentials. It provides a shared workflow for model configuration, training, fine-tuning, checkpointing, and inference. Its current force-field implementation is the Möbius family.

Möbius combines local many-body descriptors with element-conditioned residual networks. Its three architectures differ in how atomic environments exchange information: the local architecture retains scalar local features, the scalar-message architecture adds transient directional features, and the equivariant architecture retains angular states between interaction blocks. Forces and virials are obtained by differentiating the total energy.

Three trained Möbius models cover 89 elements and contain 2.98, 3.33, and 5.00 million parameters. After OMat24 pretraining and sAlex/MPTrj fine-tuning, the three architectures achieve Matbench Discovery composite scores of 0.6804, 0.7724, and 0.8410, respectively. At a common size of 18,522 Fe atoms, their native GPUMD throughputs are 2.004e+06, 2.977e+05, and 3.238e+04 atom·step/s per A100 GPU.

Research purpose and relationship to existing work

RUNE provides a configuration-driven workflow that connects model definitions to training, fine-tuning, inference, and deployment. Model configurations specify descriptor channels, residual-block depth, message layers, and retained angular states, while run configurations specify data, targets, optimization, and checkpointing. The same model interface is used for energy, force, and virial prediction.

The Möbius family combines a local many-body descriptor with element-conditioned residual encoding, an atomic-energy readout, and learned communication between atomic environments. Its three architectures share the descriptor and readout while differing in their retained interaction states. The local architecture uses scalar local features, the scalar message-passing architecture contracts transient directional features within each interaction block, and the equivariant architecture propagates angular states across blocks.

Equivariant message-passing potentials such as NequIP and MACE, together with scalar-vector models such as PaiNN, provide related approaches to directional information exchange (Batzner et al. 2022; Batatia et al. 2022; Schütt et al. 2021). Möbius covers two interaction regimes within a common formulation: transient directional features are contracted inside the scalar message-passing blocks, whereas angular states are retained and updated in the equivariant blocks. The resulting models are connected to ASE inference and native GPUMD simulation through the RUNE workflow (Xu et al. 2025).

The Möbius model family

The Möbius family comprises three architectures: the local architecture (Local), the scalar message-passing architecture (MP), and the equivariant architecture (EQ). In the remainder of the report, Local, MP, and EQ denote the corresponding implementation variants.

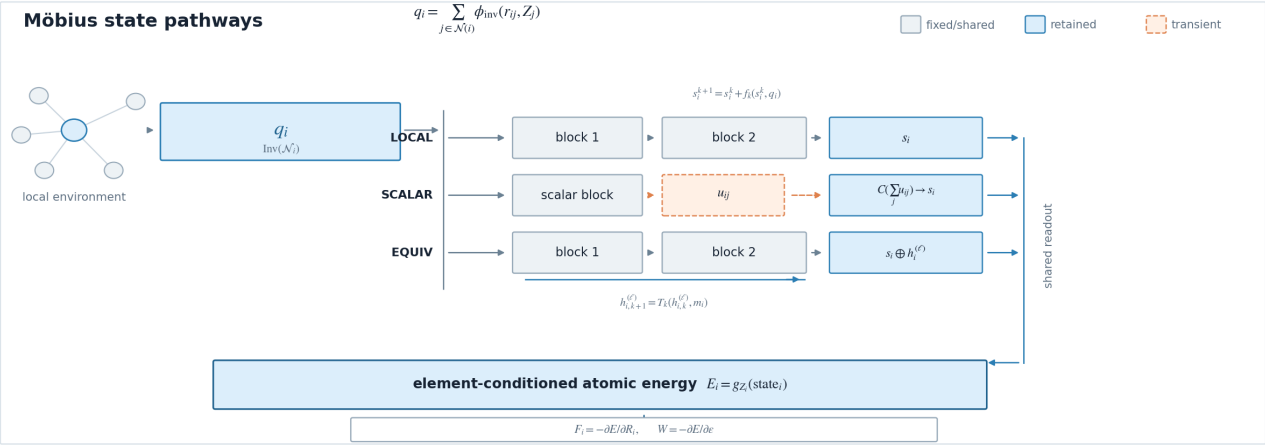


Figure 1. Möbius state pathways. The three architectures share the local invariant descriptor q_i and the element-conditioned atomic-energy readout. Local propagates only scalar states; MP forms transient directional features u_ij inside each interaction block and contracts them to a scalar update; EQ carries angular states h_i^(ell) across blocks. Forces and virials are obtained by differentiating the total energy.

The schematic separates the shared geometric representation from the state contract used by each interaction regime. The descriptor is formed once for each atomic environment, whereas the retained state determines what crosses an interaction-block boundary. Transient directional features can therefore enrich a scalar update without becoming persistent hidden channels, while EQ preserves angular channels across blocks. All three routes terminate in the same element-conditioned energy head, so their energy, force, and virial interfaces remain identical.

Local descriptors and the local architecture

Möbius uses an NEP-style local invariant descriptor as its geometric input (Fan et al. 2021, 2022). Local, MP, and EQ share this descriptor and differ in the hidden states exchanged between atoms. For a directed neighbor edge, let $\mathbf{r}_{ij} = \mathbf{R}_j + \mathbf{o}_{ij} - \mathbf{R}_i$, $r_{ij} = |\mathbf{r}_{ij}|$, and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$, where $\mathbf{o}_{ij}$ is a periodic image displacement. Element-pair-dependent radial functions and angular basis functions define

$$q^{\mathrm{r}}_{in} = \sum_{j \in \mathcal{N}^{\mathrm{r}}_i} g^{\mathrm{r}}_n(r_{ij}; Z_i, Z_j),$$

$$S_{in\ell m} = \sum_{j \in \mathcal{N}^{\mathrm{a}}_i} g^{\mathrm{a}}_n(r_{ij}; Z_i, Z_j) B_{\ell m}(\hat{\mathbf{r}}_{ij}), \qquad \mathbf{A}_{in\ell} = D_\ell \mathbf{S}_{in\ell}.$$

Here n labels radial channels, m labels the $2\ell+1$ real angular components, and $B_{\ell m}$ is the angular basis. The fixed map D_ℓ normalizes the angular moments so that their contractions can form rotational invariants.

The angular descriptor contains quadratic norms and selected higher-degree scalar contractions of these moments. For an order tuple $\lambda = (\ell_1, \ldots, \ell_p)$, write

$$\mathcal{I}_\lambda(\mathbf{A}_{in}) = \sum_{m_1,\ldots,m_p} C^\lambda_{m_1 \cdots m_p} \prod_{a=1}^{p} A_{in\ell_a m_a},$$

$$\mathbf{q}_i = \left[\{q^{\mathrm{r}}_{in}\}_n, \{\mathcal{I}_\lambda(\mathbf{A}_{in})\}_{\lambda \in \Lambda, n} \right],$$

The tensors C^λ are fixed contraction coefficients expressed in the normalized basis. The descriptor construction is shared across the three architectures. The architectural differences arise after descriptor formation, in the scalar states and directional information exchanged between atomic environments.

A componentwise scale, affine projection, and element embedding initialize a scalar state. A configurable stack of pre-normalized residual blocks transforms this state. The local architecture sends the resulting representation directly to the element-conditioned energy head, whereas the scalar message-passing and equivariant architectures pass it through their respective interaction blocks. The number and width of the residual and interaction blocks are specified by the model configuration. All three architectures use the same atomic-energy readout,

$$
\begin{aligned}
\mathbf{u}_i^0 &= P(\mathbf{d} \odot \mathbf{q}_i) + \mathbf{e}_{Z_i}, \\
\mathbf{u}_i^{b+1} &= \mathbf{u}_i^b + \mathrm{FFN}_b\Big(\mathcal{N}_b(\mathbf{u}_i^b)\Big), \qquad b = 0, \dots, B-1, \\
\mathbf{h}_i^0 &= \mathbf{u}_i^B, \\
E_\theta &= \sum_i \left[\mathbf{w}_{Z_i}^\mathsf{T} \mathcal{N}_{\mathrm{out}}(\mathbf{h}_i^{\mathrm{final}}) + b_{Z_i} + \varepsilon_i^{\mathrm{ZBL}}\right].
\end{aligned}
$$

Here $\odot$ denotes componentwise multiplication, $\mathcal{N}$ is RMS normalization with a learned channel gain, and each feed-forward map contains two affine layers separated by an activation. The optional ZBL term is added to the atomic energy. Hidden maps are shared across atoms, while the final linear head is element-specific. The local architecture sets $\mathbf{h}_i^{\mathrm{final}} = \mathbf{h}_i^0$ and exchanges no hidden features. Its atomic energies nevertheless depend on neighboring distances and angles through $\mathbf{q}_i$.

Scalar messages and transient vectors in the scalar message-passing architecture

For one scalar message-passing block, let $\overline{\mathbf{h}}_i = \mathcal{N}_{\mathrm{in}}(\mathbf{h}_i) \in \mathbb{R}^d$, and let $\pi(c)$ assign scalar channel c to one of G gate groups. Radial basis values $\mathbf{f}(r_{ij})$ and an ordered element-pair embedding $\mathbf{p}_{Z_i Z_j}$ define the edge features and gates,

$$
\begin{aligned}
\boldsymbol{\xi}_{ij} &= \Phi_e\big([\mathbf{f}(r_{ij}), \mathbf{p}_{Z_i Z_j}]\big), \\
\mathbf{a}_{ij}^s &= f_c(r_{ij}) \,\mathrm{sigmoid}(G_s \boldsymbol{\xi}_{ij} + \mathbf{b}_s), \\
\mathbf{a}_{ij}^v &= f_c(r_{ij}) \,\mathrm{sigmoid}(G_v \boldsymbol{\xi}_{ij} + \mathbf{b}_v), \\
m_{ic} &= \sum_{j \in \mathcal{N}_i^{\mathrm{m}}} a_{ij,\pi(c)}^s (W_v \overline{\mathbf{h}}_j)_c.
\end{aligned}
$$

Here Φ_e is an affine map followed by an activation, f_c is the message cutoff function, and $\mathcal{N}_i^{\mathrm{m}}$ is the message neighborhood. The scalar gates are grouped so that channels assigned to the same group share one edge-dependent gate. A separate branch constructs C_v transient vector channels,

$$
\begin{aligned}
\mathbf{v}_{ia} &= \sum_{j \in \mathcal{N}_i^{\mathrm{m}}} a_{ija}^v (W_s \overline{\mathbf{h}}_j)_a \widehat{\mathbf{r}}_{ij}, \qquad a = 1, \dots, C_v, \\
t_{ia} &= \left\| \sum_{b=1}^{C_v} M_{ab} \mathbf{v}_{ib} \right\|^2, \\
\mathbf{h}_i^+ &= \mathbf{h}_i + \Psi([\mathcal{N}_m(\mathbf{m}_i), \mathbf{t}_i]).
\end{aligned}
$$

The matrix $M \in \mathbb{R}^{C_v \times C_v}$ mixes vector channels while acting identically on the three spatial components. The squared norms t_{ia} are therefore rotational invariants. The two-layer map Ψ combines the normalized scalar messages with these invariant contractions and returns a scalar residual update.

The transient vectors are rebuilt within each interaction block and are not retained as directional hidden states. Only the updated scalar state $\mathbf{h}_i^+$ is passed to the next block. Directional information therefore influences later computation through block-local invariant contractions, whereas the equivariant architecture retains angular states across blocks.

Persistent angular states in the equivariant architecture

The equivariant architecture retains both scalar features and angular states $H_i^\ell \in \mathbb{R}^{C \times (2\ell+1)}$ for a configurable set $\mathcal{L}$ of angular orders. For each retained order, a bias-free map $P_\ell \in \mathbb{R}^{C \times n_{\mathrm{a}}}$ projects the n_{a} angular radial channels. The edge network additionally receives scalar contexts and invariant probes from both endpoints,

$$
\begin{aligned}
H_i^{\ell,0} &= P_\ell A_i^\ell, \qquad A_i^\ell \in \mathbb{R}^{n_{\mathrm{a}}\times(2\ell+1)},\\
\mathbf{p}_{ij}^{i,\ell} &= H_i^\ell \mathbf{Y}_\ell(\widehat{\mathbf{r}}_{ij}),\\
\mathbf{p}_{ij}^{j,\ell} &= H_j^\ell \mathbf{Y}_\ell(-\widehat{\mathbf{r}}_{ij}) = (-1)^\ell H_j^\ell \mathbf{Y}_\ell(\widehat{\mathbf{r}}_{ij}),\\
\boldsymbol{\xi}_{ij} &= \Phi_e\Big([\mathbf{f}(r_{ij}), \mathbf{p}_{Z_iZ_j}, C_{\mathrm{c}}\overline{\mathbf{h}}_i, C_{\mathrm{n}}\overline{\mathbf{h}}_j, \{\mathbf{p}_{ij}^{i,\ell}, \mathbf{p}_{ij}^{j,\ell}\}_{\ell\in\mathcal{L}}]\Big),\\
\{\boldsymbol{\tau}_{ij}^\ell, \boldsymbol{\eta}_{ij}^\ell\}_{\ell\in\mathcal{L}} &= \tanh(G_{\mathrm{ang}}\boldsymbol{\xi}_{ij} + \mathbf{b}_{\mathrm{ang}}).
\end{aligned}
$$

The bias-free context maps C_{c} and C_{n} are shared across all edges of a block. The last output is reshaped into two C-component gates for each retained order. The scalar message $\mathbf{m}_i$ uses the grouped sigmoid gate defined for the scalar message-passing architecture, evaluated with the equivariant edge features. The signed angular gates control transport and injection,

$$
\begin{aligned}
H_i^{\ell,+} &= H_i^\ell + \sum_{j\in\mathcal{N}_i^{\mathrm{m}}} f_c(r_{ij})\left[\mathrm{diag}(\boldsymbol{\tau}_{ij}^\ell) W_\ell H_j^\ell + \boldsymbol{\eta}_{ij}^\ell \mathbf{Y}_\ell(\widehat{\mathbf{r}}_{ij})^{\mathsf{T}}\right],\\
t_{ia}^\ell &= \sum_m (H_{iam}^{\ell,+})^2,\\
\mathbf{h}_i^+ &= \mathbf{h}_i + \Psi\Big([\mathcal{N}_m(\mathbf{m}_i), \{\mathbf{t}_i^\ell\}_{\ell\in\mathcal{L}}]\Big).
\end{aligned}
$$

Each $W_\ell \in \mathbb{R}^{C\times C}$ mixes channels within one angular order, and each gate is shared across that order's angular components. The scalar update uses norms of the updated angular states. Both $\mathbf{h}_i^+$ and $\{H_i^{\ell,+}\}_{\ell\in\mathcal{L}}$ persist between blocks. The angular branch has no componentwise angular activation or bias. Different orders enter the common invariant gate network, while angular transport uses same-order channel maps rather than general tensor products. Thus Local, MP, and EQ specify different retained representations, not consecutive stages of one network. Their scalar energies are invariant under rigid translations and rotations, and differentiation gives forces with the corresponding vector transformation law.

Forces and virials

The same energy model provides energies, forces, and virials. For derivative evaluation, RUNE treats the descriptor $\mathbf{q}$, the raw angular moments $\mathbf{S}$ used by the equivariant initialization, and the explicit edge vectors $\mathbf{r}$ as separate inputs to the energy network $\mathcal{E}_\theta(\mathbf{q}, \mathbf{S}, \mathbf{r})$. Let $J_u = \partial u/\partial \mathbf{R}$ denote the Jacobian of each geometric input with respect to the flattened atomic coordinates. The total derivative is

$$
\begin{aligned}
\nabla_{\mathbf{R}} E_\theta &= J_q^{\mathsf{T}} \nabla_q \mathcal{E}_\theta + J_S^{\mathsf{T}} (\nabla_S \mathcal{E}_\theta)_{q,r} + J_r^{\mathsf{T}} (\nabla_r \mathcal{E}_\theta)_{q,S},\\
\mathbf{F}_i &= -\nabla_{\mathbf{R}_i} E_\theta, \qquad W_{\alpha\beta} = -\left.\frac{\partial E_\theta(\mathcal{X}_\epsilon)}{\partial \epsilon_{\alpha\beta}}\right|_{\epsilon=0}, \qquad \boldsymbol{\sigma} = -\mathbf{W}/V.
\end{aligned}
$$

The first term includes the moment contractions already represented in $\mathbf{q}$. The second term is the independent angular-initialization path, including the normalization $\mathbf{A} = D\mathbf{S}$. The last term accounts for the explicit geometry used by the message-passing blocks and the optional ZBL contribution.

Network partial derivatives with respect to these inputs are obtained by automatic differentiation. RUNE then contracts the descriptor, angular-moment, and direct-edge geometry derivatives analytically. The strained configuration $\mathcal{X}_\epsilon$ transforms both the atomic positions and the periodic cell by $I + \epsilon$, with V denoting the cell volume. RUNE returns the nine components of the total virial for each structure, and its ASE interface converts the virial to stress using the sign convention $\boldsymbol{\sigma} = -\mathbf{W}/V$.

Möbius-OAM models: training and evaluation

Trained models and optimization

The Möbius-OAM training runs combine OMat24 pretraining with fine-tuning on sAlex and MPTrj. OMat24 and the sAlex subset are described in the OMat24 dataset work (Barroso-Luque et al. 2026), and MPTrj originates from the CHGNet data collection (Deng et al. 2023). The three models in Table 1 encode the same ordered set of 89 elements and use four local residual blocks. The reported checkpoints are the weights obtained after one mixed fine-tuning epoch.

Table 1. Trained Möbius-OAM configurations. FF is the local feed-forward width.

Model	Parameters	Hidden / FF	Message state	Pretraining
Local	2,977,803	128 / 256	No messages	6 epochs
MP	3,327,499	128 / 256	2 blocks; 8 transient vector channels	6 epochs
EQ	5,001,403	256 / 512	2 blocks; orders 1–4, 32 channels/order	6 epochs

Pretraining traversed 100,824,585 OMat24 structures per epoch using AdamW, an initial learning rate of 0.001, and a cosine minimum of 0.00001. It used 16 GPUs and 16 structures per rank.

Fine-tuning used a new AdamW optimizer with a constant learning rate of 0.0001. It presented the 10,447,765 sAlex structures once and the 1,580,395 MPTrj structures eight times, totaling 23,090,925 presentations in one mixed epoch. Eight GPUs each processed 32 structures per batch. Both stages used an effective batch of 256 structures, gradient accumulation of one, seed 42, weight decay 0.0001, and energy/force/virial loss weights of 1/1/0.1.

The fitting objective is

$$\mathcal{L} = \lambda_E \,\mathrm{MAE}(E/N) + \lambda_F \,\mathrm{MAE}(F) + \lambda_W \,\mathrm{MAE}(W/N).$$

Energy errors are normalized per structure by its atom count. Force errors are pooled over Cartesian components; the virial term uses six independent components per structure and is normalized per atom. Energy and virial label masks select contributing structures. Distributed training normalizes each term by the global number of available labels across the accumulation window.

Matbench Discovery

Matbench Discovery evaluates the use of potentials in materials discovery (Riebesell et al. 2025). The recorded protocol covers 256,963 WBM structures and 103 phonon materials. Predictions used float32. FIRE with FrechetCellFilter relaxed structures to a 0.02 eV/Å force threshold or a 500-step limit. Thermal conductivity used 300 K, 0.01 Å displacements, no non-analytical correction, and the complete benchmark supercells and q meshes.

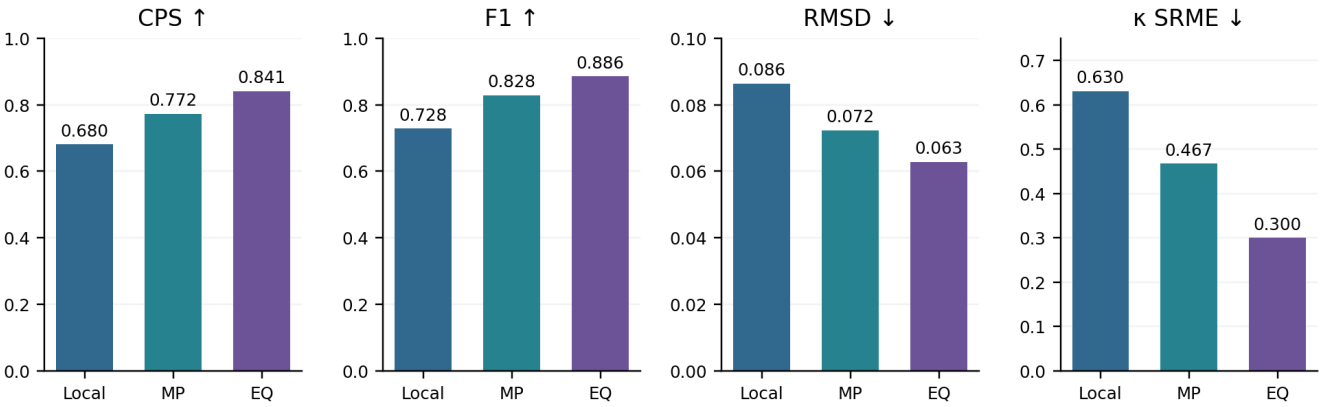


Figure 2. Matbench Discovery results for the trained Möbius-OAM models. Higher composite performance score (CPS) and F1, and lower geometry RMSD and thermal-conductivity SRME, are favorable.

Table 2. Scores and numerical outcomes. Geometry RMSD is unitless. Unconverged relaxations remain in the evaluation; unmatched geometries and imaginary phonon modes receive the recorded protocol penalties.

Metric or outcome	Local	MP	EQ
CPS	0.680395	0.772360	0.840992
F1, unique prototypes	0.728017	0.828043	0.885614
Geometry RMSD	0.086342	0.072268	0.062821
Thermal-conductivity SRME	0.630262	0.467413	0.299675
Unconverged relaxations	296	319	250
Unmatched geometries, penalized	17,325	14,621	13,055
Materials with imaginary modes, penalized	3	0	1

CPS and F1 increase from Local to MP to EQ, while geometry RMSD and thermal-conductivity SRME decrease. The outcome counts qualify this pattern: MP has more unconverged relaxations than Local, whereas EQ has one material with imaginary modes. Prediction, geometry-analysis, and conductivity execution failure counts were zero. Successful execution, relaxation convergence, and phonon stability therefore remain distinct outcomes.

Computational performance in GPUMD

The final Local, MP, and EQ weights were benchmarked in separate single-GPU tasks on NVIDIA A100-SXM4-80GB devices.

Before timing, each exported model was compared with RUNE at identical coordinates in a displaced 250-atom bcc Fe cell. The largest absolute differences across the recorded checks were 5.33e-15 eV in atomic energy, 2.06e-10 eV/Å in a force component, and 2.98e-10 eV in a total virial component.

Throughput and GPU memory measurements

Performance was measured on periodic bcc Fe cells with a 2.87 Å lattice constant, an initial temperature of 300 K, and NVE dynamics with a 0.5 fs time step. The scan began at 432 atoms, and each completed cell size was evaluated in three runs. Throughput was calculated from the timed MD segment as

$$Q = \frac{Nn}{T},$$

where N is the atom count, n is the number of timed steps, and T is the elapsed time. Reported values are medians over the three repetitions, and the error bars span the first to third quartiles.

The upper panel of Fig. 3 compares the throughput of the three models across seven common system sizes, up to 18,522 atoms. At this size, Local, MP and EQ achieve throughputs of 2.004e+06, 2.977e+05 and 3.238e+04 atom·step s(^{-1}), respectively. The lower panel shows the peak GPU memory usage at each size, measured using `nvidia-smi` at 200-ms intervals over three independent runs. Extended memory tests reached maximum successfully completed system sizes of 1,024,000 atoms for Local, 109,744 for MP and 27,648 for EQ. All tested configurations completed without native out-of-memory errors, establishing their maximum demonstrated operating scales on the 80-GiB A100 GPUs.

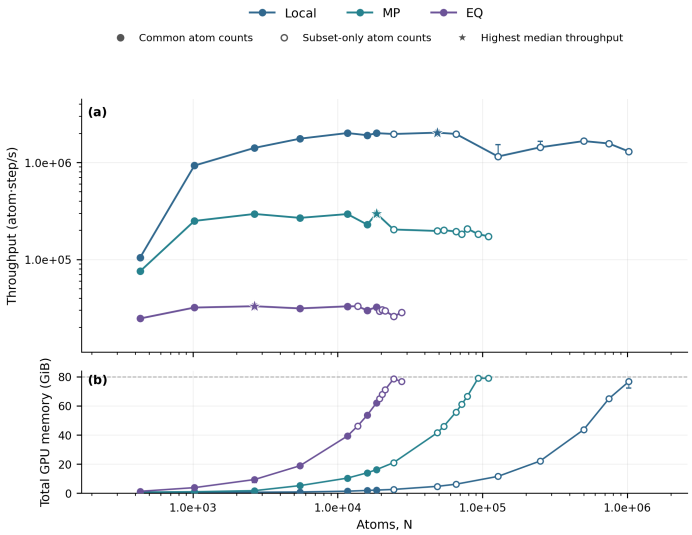


Figure 3. Native GPUMD throughput and sampled total GPU memory for the final Möbius-OAM models. The upper panel shows throughput medians with first-to-third-quartile error bars; the lower panel shows sampled total GPU memory peaks. Filled circles denote common atom counts, open circles denote subset-only atom counts, and stars mark the highest measured median throughput for each model.

Saturated inference comparison

DPA4 reports saturated inference throughput from the ASE calculator on a single NVIDIA H20, whereas the Möbius reference points are the highest median throughputs from native GPUMD full-MD measurements on a single A100. The two measurements use different hardware, software paths, and timing scopes, so Figure 4 provides a reference comparison rather than a direct ranking. The Möbius reference maxima are 2.027e+03, 2.977e+02, and 3.301e+01 atoms/ms for Local, MP, and EQ, respectively.

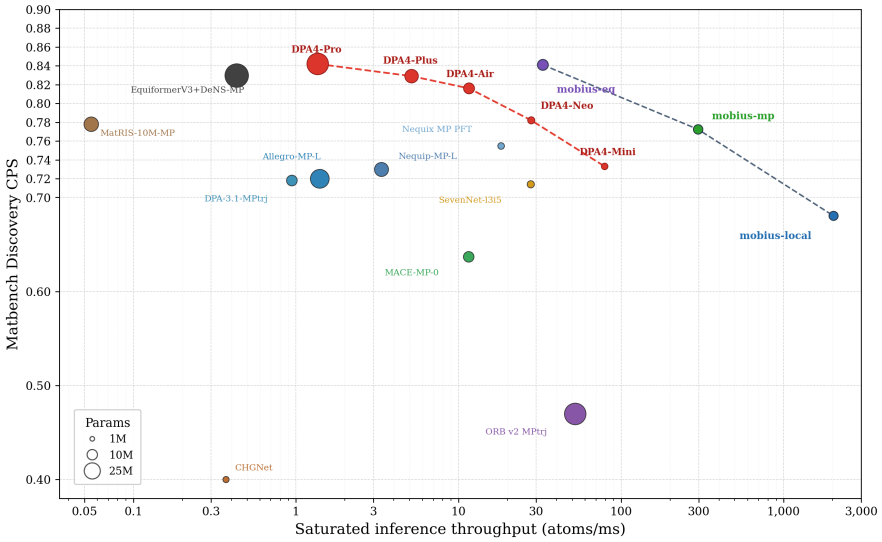


Figure 4. Accuracy–throughput comparison using Matbench CPS and saturated inference throughput. DPA4 values are reported in arXiv:2606.02419v4. Möbius Local, MP, and EQ are shown as native GPUMD A100 reference maxima. The two series use different benchmark protocols.

Discussion

The results show a consistent tradeoff between model expressiveness and computational cost. From Local to MP to EQ, the Matbench Discovery scores and geometry-related metrics improve, while native GPUMD throughput decreases. This ordering is consistent with the retained interaction states: Local uses scalar local features without hidden-feature exchange, MP introduces transient directional information that is contracted within each block, and EQ retains angular states across blocks.

RUNE provides the workflow needed to study this tradeoff systematically. Its configuration-driven model and run definitions connect descriptor choices and interaction states to training, fine-tuning, energy, force, and virial evaluation, as well as deployment, through a shared interface. This makes Local, MP, and EQ comparable within one model family while keeping the same downstream prediction and simulation paths. The value of RUNE therefore extends beyond a single potential: it provides a reproducible structure for developing and evaluating modular potential families.

The present results also expose the role of controlled ablations in interpreting modular potential families. Because the models vary in parameter count, hidden width, and training configuration, the observed performance ordering combines representation choice with model capacity and training history. A controlled series that holds the descriptor, element set, datasets, optimization schedule, random seed, and parameter budget fixed while varying message depth, transient vector channels, angular orders, or channel width would help isolate these effects. Evaluating such a series with common energy, force, virial, discovery, throughput, and memory metrics would reveal how each architectural choice moves the cost–accuracy frontier. This is where RUNE's configuration-driven workflow has particular value: the same model, training, evaluation, and deployment interfaces can support these controlled comparisons.

AI assistance

Generative AI tools assisted with language editing and organization of this technical report. Model descriptions, equations, and numerical results were checked against the RUNE implementation, recorded configurations, and benchmark outputs.

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